

What a valid test of the shell looks like

HCSA shell test plan, Rev B. Manos Theofanous, ENGN 1972 with Prof. Rick Fleeter. 25 September 2026. For review by Prof. David Henann. Hand calculations; the finite element model replaces every number here.

What changed since the 22 September draft

- Every hand number was re-derived and checks out; the plate formula now says what a is (the pane altitude, 1,839 mm).
- T1 is now a size series (75, 150 and 225 mm slices), after Tengxiao Li's advice to test a few sizes and see when the response stops changing.
- The T1 grips are designed: the flange and seat lip run out to 200 mm and a steel pull block bolts between them, so the load goes in through the same parts that carry it in the panel.
- The ribs are spokes: the hub has no support, so each rib carries its load all the way to the frame corner. That makes the Rev C rib far weaker than the 22 Sep note said (about 2,900 MPa at 1 atm, not 720). The T2 rib is hand-sized as a spoke and stays elastic to 2 atm.
- New: the shell-scale question Prof. Fleeter raised on 24 September. Where does the in-plane load go between seams, and what is the smallest section that shows it? (Sheet 6.)
- The joint is simpler (Rev S): two frames bolted face to face through one row of 5/16 in bolts at 75 mm, two O-rings in straight grooves and the RTV bead in Frame B, Frame A flat. The tongue and groove, key boss, conduit, racetrack ends and guide pins are gone (Sheet 1).
- The whole shell is now modelled: 32 panels, 90 seams, every bolt, nut, washer, O-ring and bead, no two bodies overlapping. With 30 bolts per edge the end nuts of two seams collide at every pentagon corner, so each edge has 29, 75 mm from each vertex (TRUTH says about 30).
- The T2 rig is hexagonal, and the membrane has its own clamp ring so its seal does not unload when the panel lifts (Sheet 3).
- All test articles, both rigs and the whole shell are CAD (STEP, glTF, STL), including a 550 mm joint article for a dry fit and a leak test. A separate PDF has 2D and 3D sketches and renders of all of them.

The short answer

- The load that matters is cabin pressure: 1 atm pushing out on every panel, 1.33 MN per hexagon. This is a bathysphere in reverse. The shell as a whole is in tension (the panels, ribs and frames still bend locally), and it fails by breaking, leaking or tearing a seam, not by buckling. A valid test has to push from inside.
- A flat panel can't carry pressure the way a sphere does. It bends onto its frame, and the frames pass the load around the shell as tension across every seam: 277 kN per metre, which is $p R / 2$. If the panels carry their in-plane pull, that is what the bolts and gaskets see: 20.7 kN per bolt, almost three times the 7.4 kN in TRUTH. It is an upper bound; the real value depends on the shell-scale load path below.
- So one test is not enough. The minimum set is a single panel under pressure (the face bending onto the frame and ribs) plus a full-size slice of the seam pulled apart (the bolts and seal under the real seam tension). Each covers the other's gap.
- The panel test uses aluminium panes instead of glass. They bend almost like the glass, don't shatter, and can go past 1 atm. Glass strength is measured separately on small coupons, because the glass allowable is the number we understand least.
- The test doesn't have to be the flight panel. It has to be something the model can predict. If the model matches the test, we trust the model for the real shell.

- One question none of these tests answers: how the shell shares the in-plane load between the panels, the seams as beams and the seams as a net. By hand the split changes the bolt load by a factor of up to eight (35 to 277 kN/m across the joint) and the frame and corner loads by several. I propose finding it with an FEA size study first (Sheet 6).

The tests, smallest first

Test	What it proves	Size and load	Where and when
T0 glass coupons (Sheet 4)	Strength of our glass and its scatter (Weibull), so the pane allowable is measured	30+ discs per glass, ring-on-ring (ASTM C1499), about 1 to 4 kN, analysed to ASTM C1239; one set at four stress rates for slow crack growth (ASTM C1368)	Small load frame. October
T1 seam slice (Sheet 1)	Bolts, gaskets and seal under the real seam tension; gap and leak against load; prying	75, 150, 225 mm full-section slices; up to 41.5 kN at 1 atm on the 150 mm slice (upper bound)	100 kN frame. After the week 9 release
T2 one panel, 1:4 (Sheets 2, 3)	How the face bends onto the frame and ribs; stress map; the edge boundary condition	1,125 mm across, 8 mm aluminium panes; 83 kN at 1 atm, 167 kN at 2 atm	Water rig in a shop bay. Spring 2027
T3 corner node	Three panels at a corner: the corner seal and how three seams share load	Scaled or partial, after T1 and T2 correlate	Spring 2027 or later
S whole shell, 1:10 (Sheet 6)	The load split across the shell, only if the FEA size study says a cap is not enough	1.12 m closed shell under water; self-reacting; run elastic at about 0.3 atm	Spring 2027 at the earliest
T4 full-size panel	The real thing at full scale	1.33 MN at 1 atm; industrial rig	Only with a partner

The model predictions for every gauge are filed before each test. A pass is strains, deflections and gaps within 10 percent of the prediction in the linear range, a straight-line response to 1 atm that returns to zero, and in T1 no leak at 1 atm, an inner-gasket gap under 0.33 mm (a quarter of the squeeze, provisional) and no yield below proof. A miss is still useful: it says which part of the model to fix before anyone builds a full-size panel.

What the hand numbers already say

Finding	Number	What it means
Bolt load may be almost three times what TRUTH assumed	Up to 20.7 kN per bolt at 1 atm. The -10 class (about 34 kN) has 1.6 before any factor and 0.7 at ultimate; the -06 in TRUTH (about 12 kN) fails at 1 atm	Bolt size and pitch go to the top of the Rev D list. T1 measures it
The glass pane stress is high	53 MPa in a 31 mm borosilicate pane at 1 atm, simply supported	Above what annealed glass normally carries for a sustained load. T0 sets the real allowable. If the coupons come in near 150 MPa with a Weibull modulus near 7, the strength scaled to a 2 m2 pane is about 42 MPa, before any factor
The pane also fails the deflection limit	7.2 mm at the pane centre at 1 atm, simply supported, against 1/200 of the 613 mm inscribed radius = 3.1 mm	TRUTH 8.2 sized 31 mm for deflection with a circular-plate estimate. On the real triangle it only passes if the seat acts close to clamped. T2 measures the edge fixity
The Rev C ribs are far undersized	The hub has no support, so each rib is a spoke to the corner: about 2,900 MPa at 1 atm	Deeper ribs or a ring that supports them mid-span (TRUTH 9.6 lists radial plus ring). The T2 rib is 110 mm deep at 1:4. The week 6 rib study decides
The in-plane load path is open	258 kN/m in each panel's plane at every seam	An edge frame alone can't carry it (about 7,000 MPa). The seams as beams and as a net can share it, loading the corners hard, and then the bolts see far less. Needs the FEA size study

Questions for Prof. Henann

- **Is this the right minimum set?** T0, T1 and T2 plus an FEA size study, or would you test a different subassembly first, for example the corner node, given the net path on Sheet 6?
- **The shell-scale load path.** Is the hand argument on Sheet 6 right, that the in-plane load can't stay in the panels and mostly runs along the seams to the corners? If so, the bolts see much less than 277 kN/m and the corners much more. For the full-shell FEA, would you model the bolted seams with connector elements and a gap law, or with contact?
- **Boundary conditions in T2.** Do a pinned edge (on a round bar) and a sandwiched edge bracket the real edge well enough, or should one edge also carry in-plane tension? That would need a different rig.
- **Replica scaling.** Is 1:4 sound for the panel when the setting strips, beads and web width don't scale cleanly?
- **The simple joint.** Is one bolt row between the O-rings and the seat lip enough to hold the inner O-ring gap under the provisional 0.33 mm at 1 atm, or should the flange get a second row? T1 run 2 measures the prying.
- **T1 grips.** Is pulling through the extended flange and lip the right way to load the seam, or would you pull on pins through the web?
- **Facilities.** Does the School have a 100 kN frame, digital image correlation and a strain-gauge DAQ I could book, and floor space for a 700 kg water rig? Who signs off on pressure testing?

What I need to run it

Item	For	Status
Universal test frame, 100 kN, with clevis grips and a spherical seat	T1	Asking the ENGN 0310 TA (Surya Kolluri) and you
Small load frame, 5 kN, compression	T0	To ask
Strain-gauge DAQ, 16 channels; rosettes and uniaxial gauges; 2 clip gauges; 2 LVDTs or dial gauges	T1, T2	0310 lab equipment, if students can use it outside the course
Digital image correlation (optional)	T2	To ask
Shop time: 6061 plate frames, 4140 blocks; steel rig (flange ring, top ring, tub)	T1, T2	Drawings released week 9 (8 Nov)
EHS review of the water rig and the relief valve	T2	Before first pressure

How it fits the fall

- 27 Sep: requirements (15, sourced) and the reconciled load cases, in the companion note and on Sheet 5.
- October: T0 coupons ordered and run; materials from MMPDS (4 Oct); glass design strength (11 Oct); first FEA of the seam and one panel, rib depth settled (18 Oct); then the FEA size study (Sheet 6).
- 25 Oct Rev D seam; 8 Nov drawings and quotes released, go/no-go; 22 Nov the spring test plan. T1 runs as soon as parts arrive; T2 is spring work.

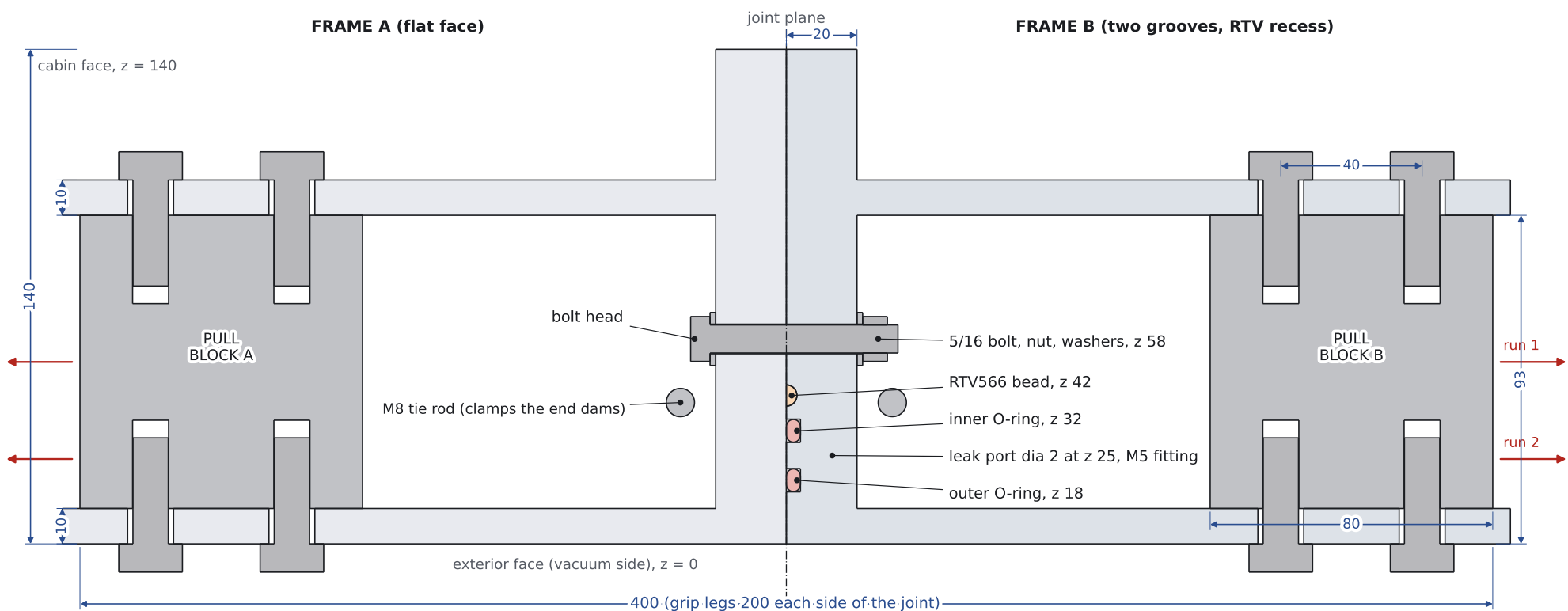
Files

CAD models of the whole shell, single panels, seams, a corner node and every test article (STEP, glTF, STL, a check report each), made by parametric scripts from one parameter file where every value has a status. Every body is a valid solid, no two overlap, and every file was reopened and checked after export. Hand calculations in handcalcs.py and verify_plate.py. Drawings and renders are in a separate PDF.

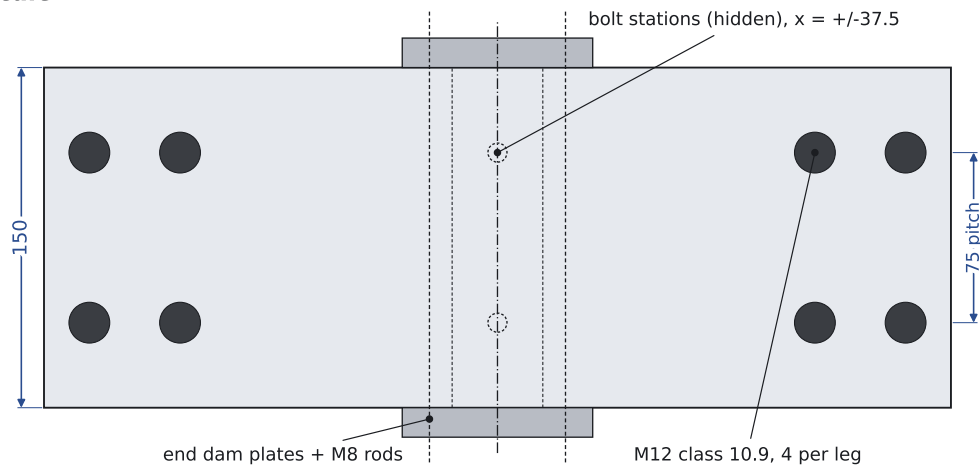
Sources

Timoshenko and Woinowsky-Krieger, Theory of Plates and Shells, 2nd ed., simply supported equilateral triangle ($w = p a^4 / 972 D$ at the centre, a the altitude; maximum moment $0.025 p a^2$ for $\nu = 0.2$). HCSA TRUTH (25 Apr 2026) App. A, 2.2 to 2.4, 8.3, 9.5, 10.2 to 10.7; the Rev C joint model and the Rev S parameter file. NASA-STD-5001B (factors of safety, to confirm), NASA-STD-5018 (glass), ASTM C1499 and C1239 (glass strength). Cabin pressure 101.325 kPa.

SECTION A-A through a bolt (x = +37.5), about 1:1.7

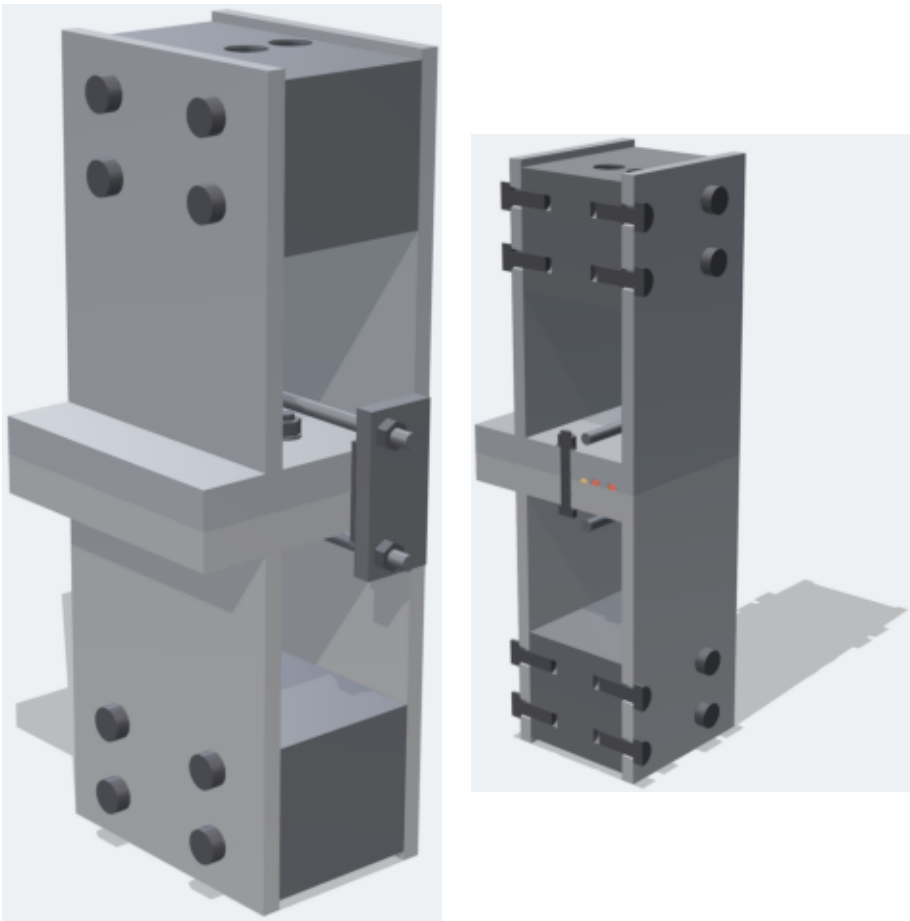


PLAN from the exterior face, same scale



How it runs

Two runs per slice. Run 1: both legs bolted, pull on z 51.5, between the flange and lip and close to the bolt line. Run 2: lip bolts removed, pull on z 24, so the load enters through the outer flange only and the joint pries open on the exterior side. Measure: load and stroke; gap opening at z 0 and z 140 with clip gauges at both ends; strain on one instrumented bolt; pressure between the O-rings through the M5 fitting. The grooves run out at the slice ends, so rubber-faced end dams, clamped by two M8 rods, close them. Charge to 1 atm and log decay while loading. Pass: inner-gasket gap under 0.33 mm at 1 atm (PROVISIONAL); no leak at 1 atm; no yield below proof; model within 10 percent.



The 150 mm slice as mounted; right, cut through a bolt.

Size series and loads (seam tension 276.7 kN/m)

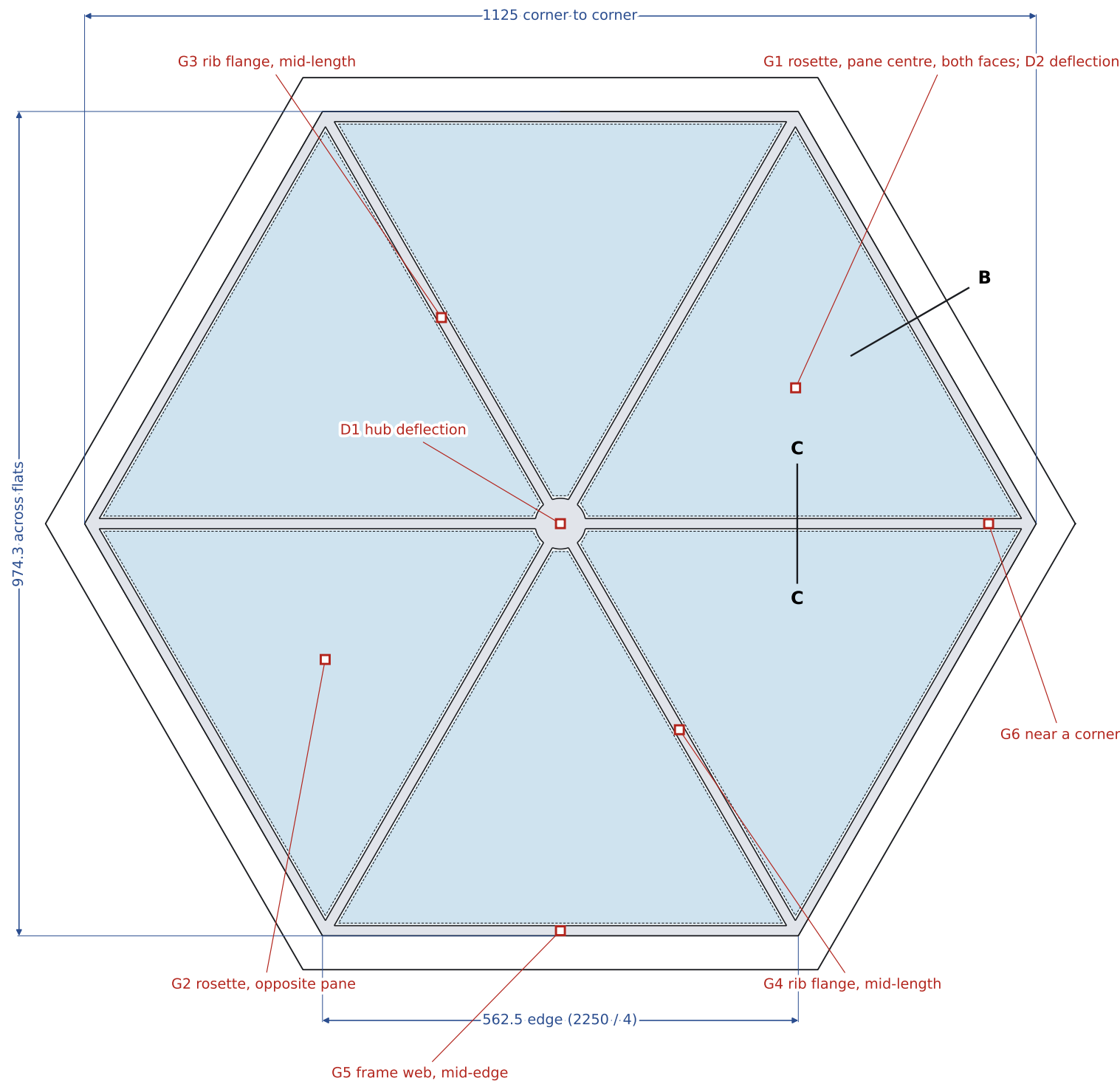
Slice	Bolts	1 atm	1.5 atm	2 atm	Run to
75 mm	1	20.7 kN	31.1 kN	41.5 kN	2.4 atm
150 mm	2	41.5 kN	62.2 kN	83.0 kN	2 atm
225 mm	3	62.2 kN	93.4 kN	124.5 kN	1.5 atm

Per bolt at 1 atm: 20.7 kN, the upper bound (Sheet 5); TRUTH says 7.4. 2.4 atm is ultimate (2.0 x LC2). 150 and 225 mm stop early to stay inside a 100 kN frame. If the per-bolt response is the same for 2 and 3 bolts, 150 mm is the working size.

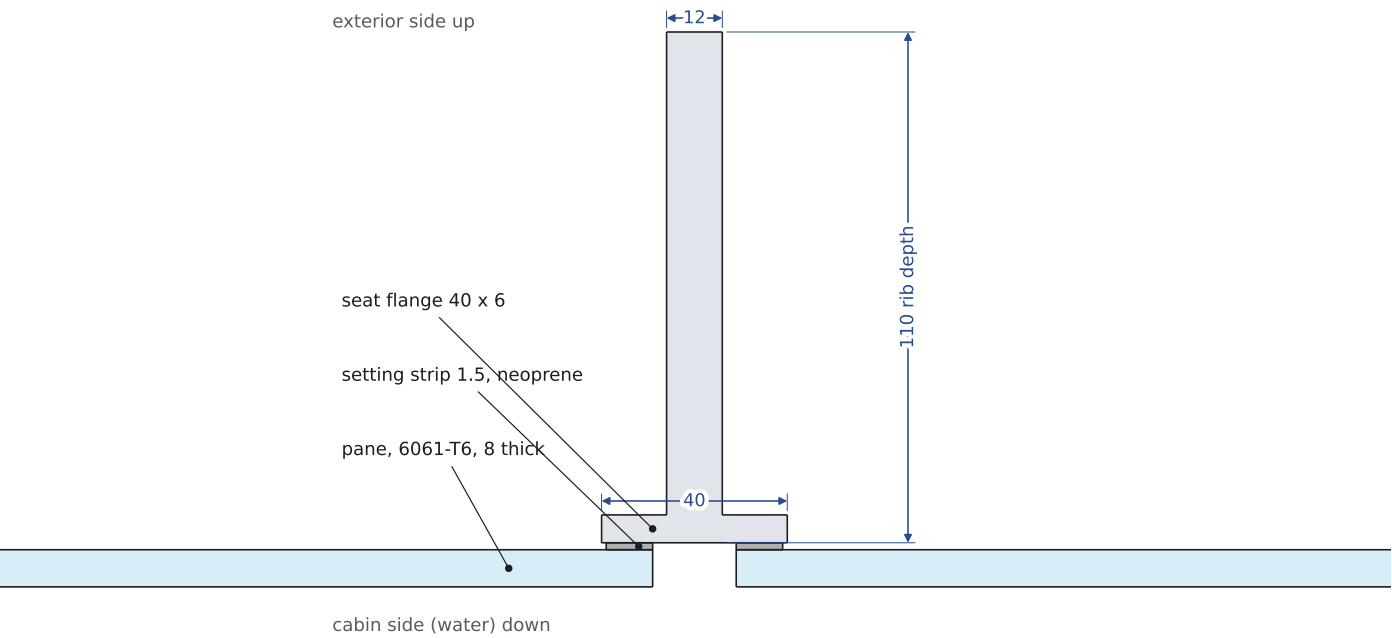
Parts

The simple joint (Rev S): flat faces, one bolt row, two straight O-ring grooves and the RTV recess in Frame B. No tongue and groove, key boss, conduit or guide pins. Frames 6061-T6, flange and lip run out to 200 mm as grips. Pull blocks 4140, M24 x 3 to a spherical seat. Models: T1_seam_slice_75/150/225mm, T1_joint_article_550mm.

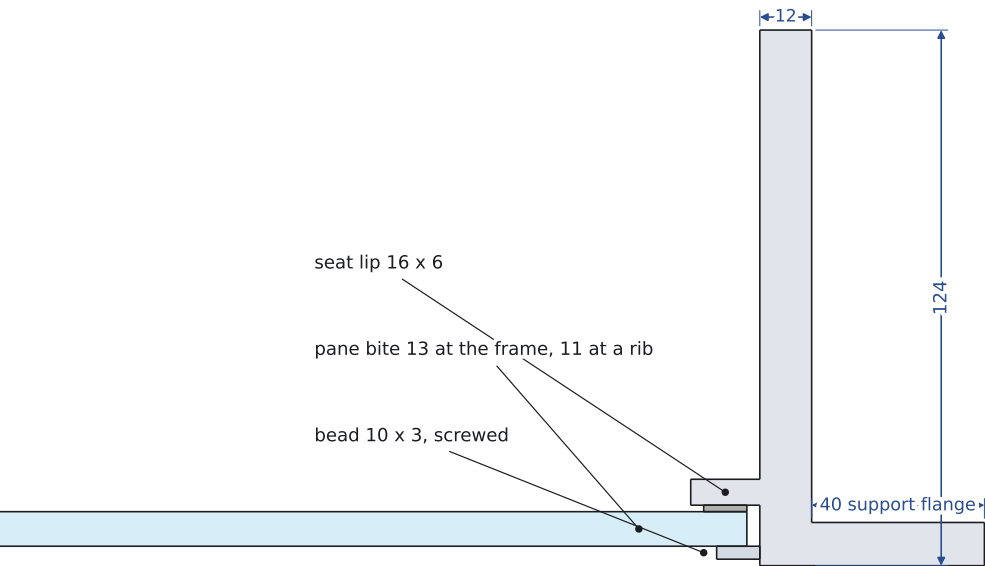
PLAN from the exterior side, about 1:6.4



SECTION C-C across a rib, about 1:1.6



SECTION B-B at a frame edge, about 1:1.6

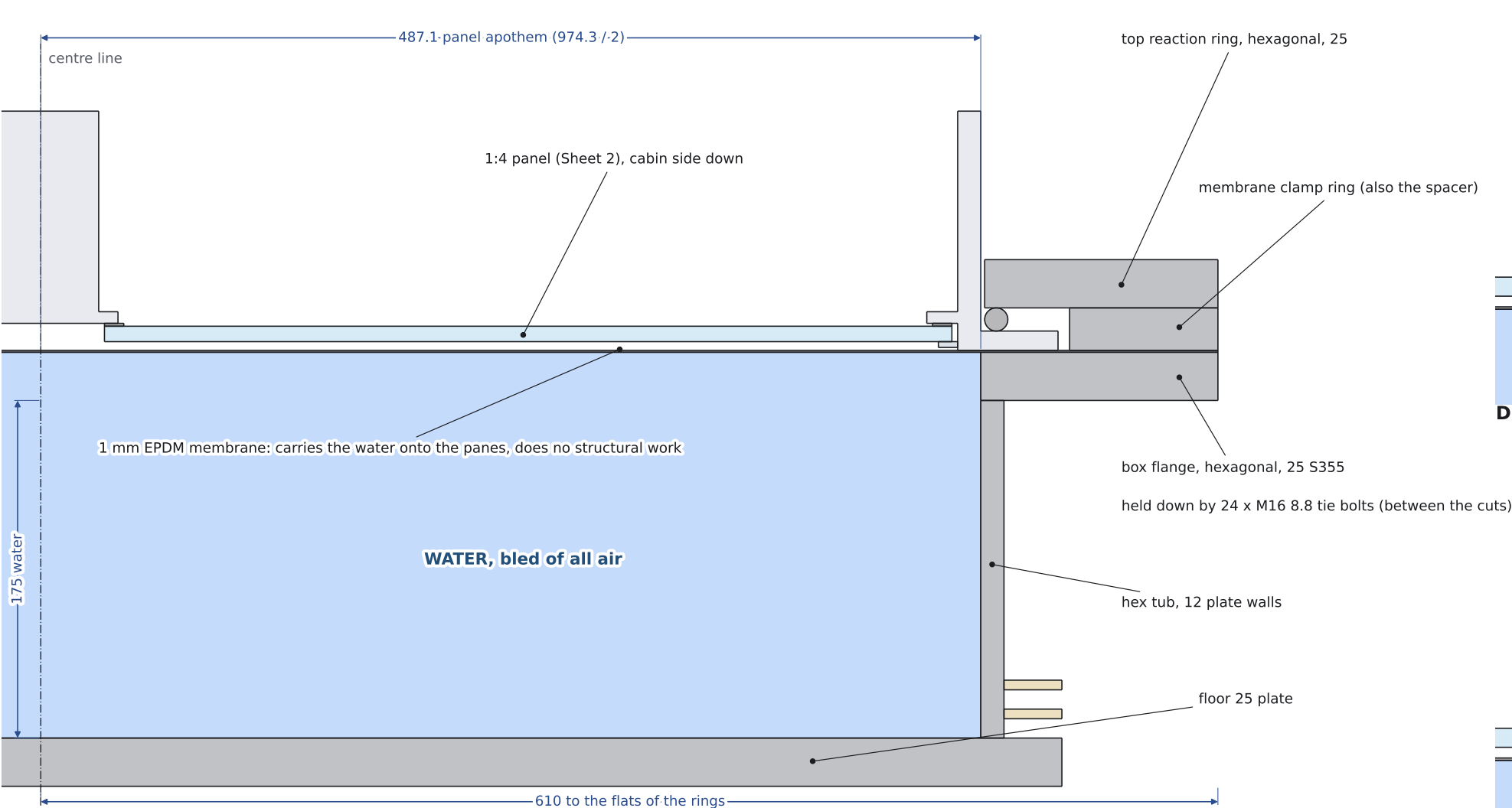


What the article is

The hub has no support, so each rib is a spoke: it carries its load out to the frame corner ($M = w L^2 / 2$). Hand-sized (PROVISIONAL) to stay under yield / 1.25 at 2 atm: $S = 28,500 \text{ mm}^3$, 91 MPa at 1 atm, 183 MPa at 2 atm. The same logic puts the Rev C flight rib near 2,900 MPa. The week 6 rib study decides the real answer (deeper ribs, or a ring that supports them). Replica scaling: every length / 4, same stress, force / 16. Not replicas: 124 mm frame depth (35 would be), 12 mm web, 40 mm flange, 8 mm pane (7.4 matches the glass), seat strips and beads. Metal 31 kg, panes 16 kg.

At 1 atm (hand calc)	Test 1:4	Flight 1:1
Pane stress, max / at centre	53 / 50 MPa	53 / 48 MPa (glass)
Pane deflection, centre	1.4 mm	7.2 mm
Rib stress at the hub	91 MPa	~2,900 MPa (Rev C rib)
Hub deflection	2.9 mm	-
Panel force	83 kN	1333 kN

HALF SECTION through the middle of two opposite edges, about 1:3



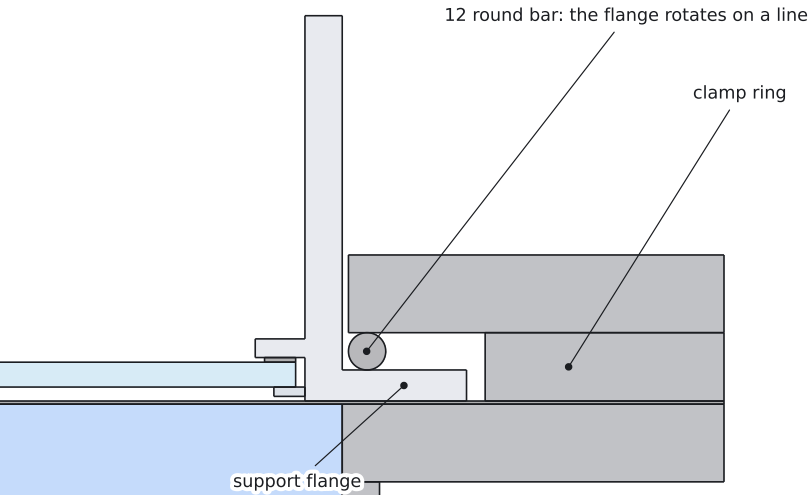
How it runs

1. Write down the model prediction for every gauge before any water goes in.
 2. Fill, bleed through the top port until no air comes out, check the relief valve lifts at 220 kPa.
 3. BC-A first. 0 to 1.0 atm in 0.25 atm steps, hold 2 min per step, read everything; unload the same way. Three times.
 4. Change to BC-B and repeat. 5. Only then 1.5 and 2.0 atm, in 0.1 atm steps above 1 atm.
- Dials read against a bridge clamped to the panel frame, so rig, ring and membrane movement drop out; one more dial on the frame edge.
- If the School has digital image correlation, one pane on the exterior side gives the full strain field.

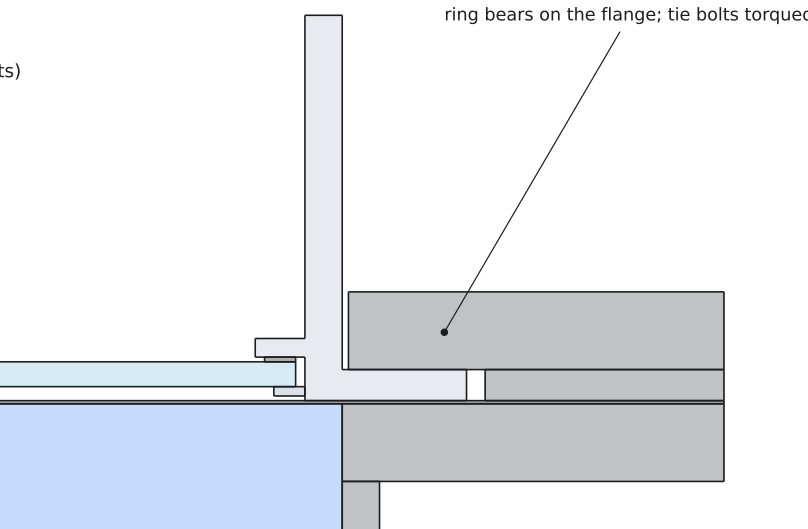
Water, not air: a failure releases almost no stored energy. The rig still goes to EHS and the shop before first pressure.

The membrane has its own clamp ring, bolted through by the tie bolts, so the seal does not unload when the panel lifts.

DETAIL BC-A, pinned edge, 1:1.5



DETAIL BC-B, clamped edge, 1:1.5



Loads on the rig

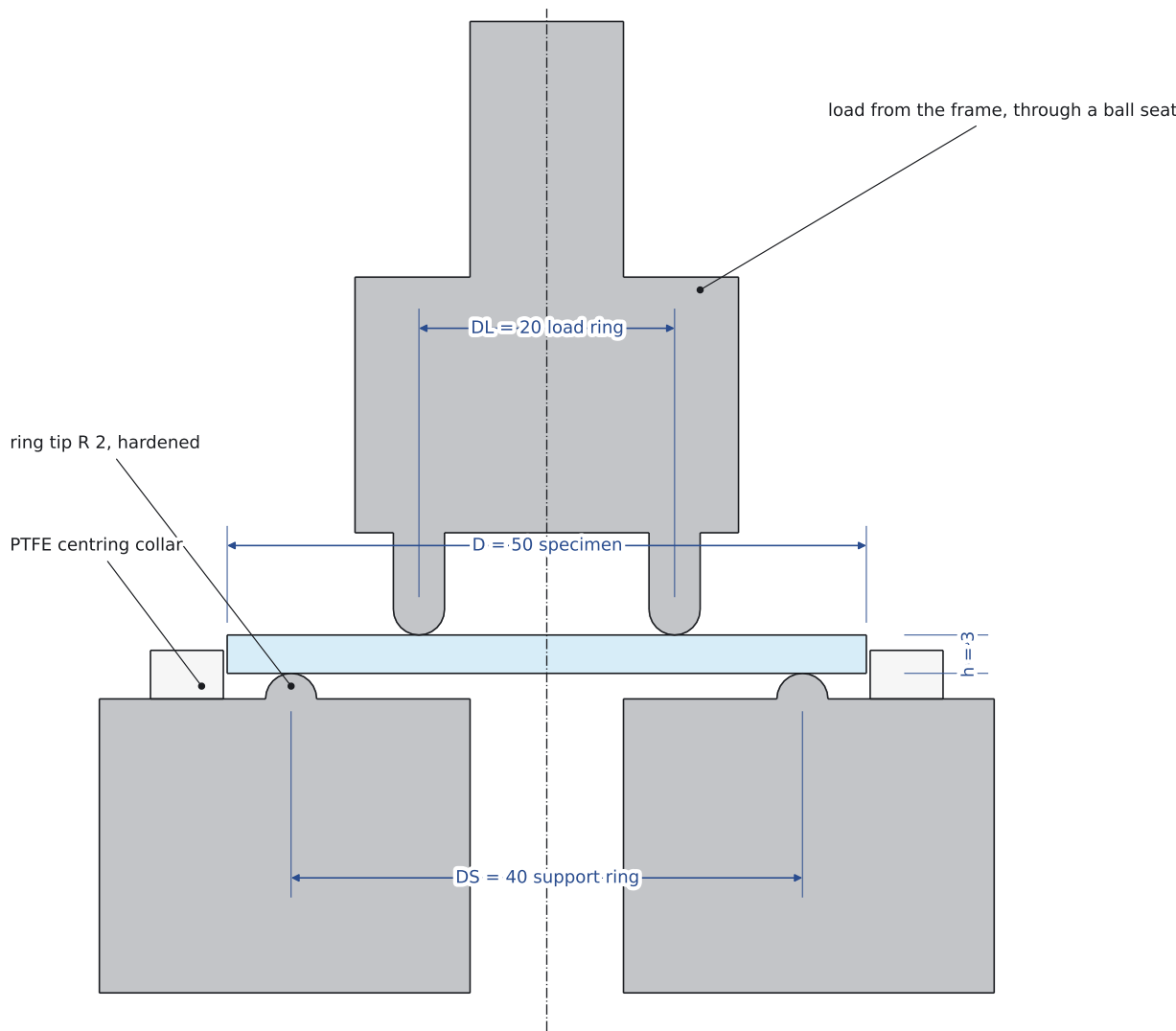
Rig loads	1 atm	2 atm
Force on the panel	83 kN	167 kN
Per metre of panel edge	25 kN/m	49 kN/m
Per M16 tie bolt (24)	3.5 kN	7 kN
Top ring bending (rough)	~20 MPa	~40 MPa
Top ring deflection (rough)	~0.1 mm	~0.2 mm

The flange, clamp ring and top ring are hexagonal, like the panel.

About 490 kg of steel and 165 kg of water; it needs a lift.

Models T2_rig_BCA_pinned and T2_rig_BCB_clamped.

SECTION through the fixture axis, 2.5:1



What T0 is for

Why: the pane stress at 1 atm is about 53 MPa (Sheet 2), and nobody has measured what our glass can carry. The allowable has to come from a measured strength distribution, not a handbook number.

What: at least 30 discs per glass and surface condition (ASTM C1239 minimum for a Weibull fit).

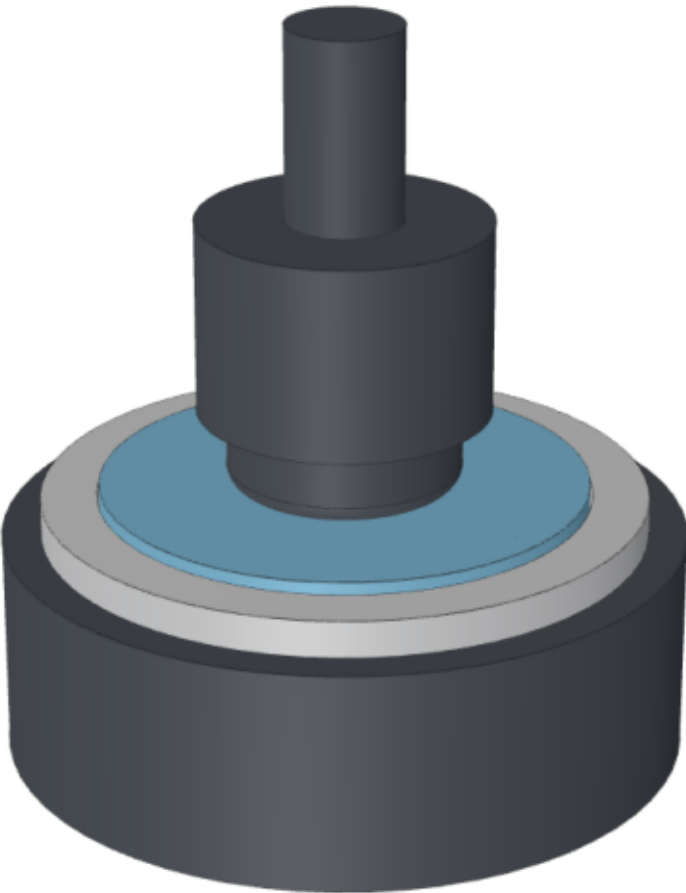
Start with as-received borosilicate and fused silica; add an abraded set, since flight glass will be scratched.

How: ring-on-ring, equibiaxial, ASTM C1499, in a small load frame. Tape the compression face so fragments stay put.

Tiny loads: about 0.9 kN at 50 MPa and 1.8 kN at 100 MPa for a 3 mm disc.

Analysis: maximum likelihood fit of m and sigma0 (ASTM C1239), then scale to the pane by effective area.

Glass also weakens under sustained load (slow crack growth), so one set runs at four stress rates (ASTM C1368).



Why the size effect decides the pane

Weibull modulus m (assumed)	5	7	10
Strength drop, coupon to a 2 m2 pane	5.9 x	3.5 x	2.4 x
150 MPa on a coupon becomes	26 MPa	42 MPa	62 MPa

Rough and conservative: full pane area against the area inside the load ring (314 mm2), not effective areas. T0 measures m.

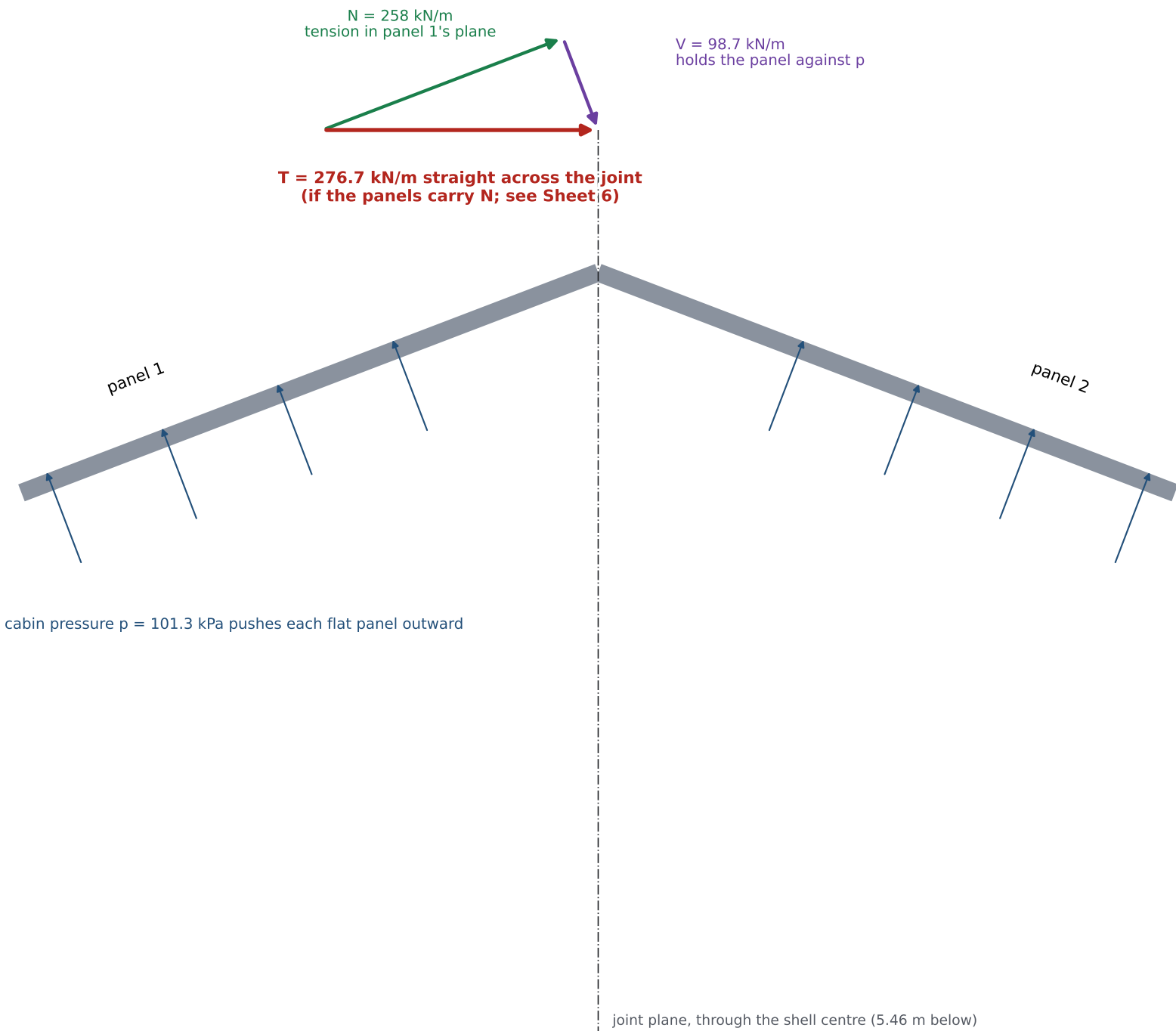
With m about 7, a coupon strength that sounds comfortable ends up below the 53 MPa the pane sees. That is the case for doing T0 first. By hand the pane also misses its deflection limit (7.2 against 3.1 mm), so both checks matter (TRUTH 9.5).

Stress in the disc (ASTM C1499):

$\sigma = 3F / (2 \pi h^2) \times [(1 - \nu)(DS^2 - DL^2) / (2 D^2) + (1 + \nu) \ln(DS / DL)]$

Fixture geometry is PROVISIONAL: check it against the C1499 limits on D, DS, DL and h before ordering discs.

EDGE VIEW of one hex-hex seam, looking along the edge (not to scale)



Each hex panel carries $p \times 13.153 \text{ m}^2 = 1.33 \text{ MN}$. Spread over 13.5 m of edge that is $V = 98.7 \text{ kN/m}$, the number in TRUTH 2.4.2. The two panels meet at 138.19 deg, so V alone cannot balance at the seam: each panel also pulls in its own plane with $N = 258 \text{ kN/m}$. V and N add to $T = V / \sin(20.905 \text{ deg}) = 276.7 \text{ kN/m}$, normal to the radial joint plane on every seam. That equals $p \text{ R} / 2$ with $R = 5.461 \text{ m}$ to the edge midpoints: the faceted shell carries the same seam load as a sphere. Per bolt at 75 mm: 20.7 kN, not 7.4.

Load cases for the seam (milestone item 2, reconciled)

Case	Seam load	Per bolt	550 mm article	Source
LC1 limit, 1 atm	276.7 kN/m	20.7 kN	152 kN	TRUTH App. A, this sheet
LC2 design, x 1.2 transient	332.0 kN/m	24.9 kN	183 kN	TRUTH 2.4.2 (docking ~20 %)
LC3 proof, 1.5 x LC2	498.0 kN/m	37.3 kN	274 kN	placeholder, NASA-STD-5001B
LC4 ultimate, 2.0 x LC2	664.0 kN/m	49.8 kN	365 kN	TRUTH 2.4.2 prototype factor
LC5 leak check	1 atm between gaskets	-	-	TRUTH 2.4.5
LC6 thermal, -40 to +120 C	range, 90 min orbit	-	-	TRUTH 2.4.2, FEA later
Old basis (V only)	98.7 kN/m	7.4 kN	54.3 kN	TRUTH 10.3, superseded

The 99 and 282 kN/m in the March plan were both real. 99 is the V component; about 277 is what the bolts carry if the panels carry N. That is the upper bound. If the in-plane load runs along the seams instead (Sheet 6), the pull across the joint drops toward $V \sin 20.9 \text{ deg} = 35 \text{ kN/m}$ and the frames carry about 1 MN along the seam. So the bolt load lies between 35 and 277 kN/m, and the load path decides where. LC1 to LC4 use the upper bound.

At the bound, the 550 mm article is a 150 to 365 kN test, too much for a 100 kN frame: T1 uses short slices, and the full article gets the dry fit and leak test.

Against the -10 bolt (about 34 kN, shank area, 689 MPa; the NASM rated strength replaces this in item 7): 1.6 at LC1, 1.4 at LC2, 0.9 at LC3, 0.7 at LC4. The -06 in TRUTH (about 12 kN) fails at LC1. Prying is not included; T1 run 2 measures it.

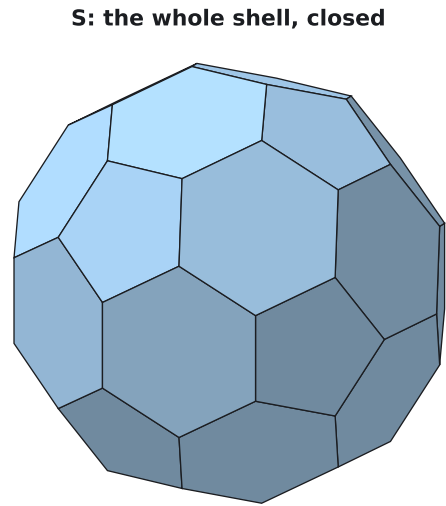
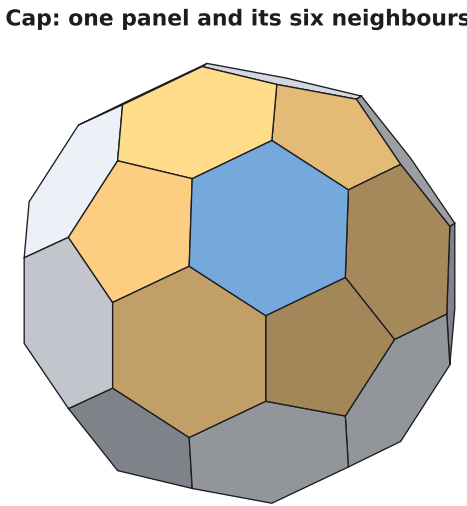
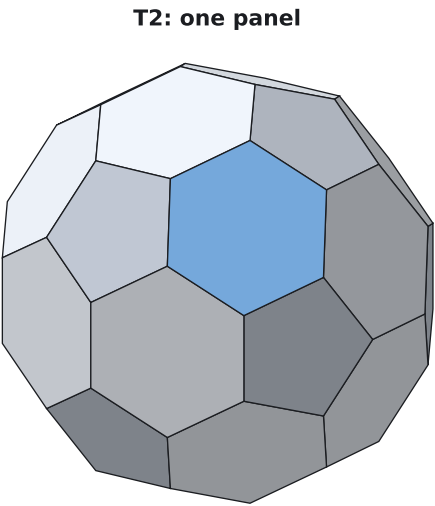
Sheet 5. What cabin pressure does at one seam, and the load cases

Hand calculation, uniform average along the edge. The finite element model replaces it.

HCSA shell test plan, Rev B
M. Theofanous, Brown ENGN 1972

25 Sep 2026
Units: mm

Sheet 5 of 6
Third angle



The question

A flat panel cannot carry pressure the way a sphere does. At every seam the panels need the 258 kN/m in-plane pull of Sheet 5, and something has to carry it from one seam to the next. The glass floats on strips, so it is not the panes. There are three candidates:

- A. The panel as a membrane. The frame and ribs carry N in the panel's own plane. TRUTH 2.3 assumes this ("the perimeter carries the force around the hex edges as tension"). But a 2.25 m edge frame loaded in its plane by 258 kN/m bends between the corners: about 109 kN m per edge if the corners are fixed, roughly 7,000 MPa in the Rev C frame. This path cannot carry it as drawn.
- B. The seam as a beam. The two frames together span corner to corner and carry the net radial push, 184 kN/m: about 117 kN m per seam, simply supported: about 690 MPa on the two Rev C frames. Much stiffer than A, still 2.5 times over yield.
- C. The seams as a net. The 90 seams act as a geodesic net in tension: about 1,025 kN along each seam, 126 MPa on two frames, and three of those meet at every corner. C is not on its own: the load has to reach the corners first, through B or the ribs.

My hypothesis is that the shell ends up mostly on B and C, with the corners heavily loaded. That also sets the bolt load: from 277 kN/m across the joint if A carries it, down toward 35 kN/m if B and C do. The split has to be found before the joint is sized for good. None of T0, T1 or T2 can find it: T2 has no neighbours, and T1 has no panel and never sees the 1 MN along the seam.

How I propose to find the minimum section

- 1. FEA size study first (fall, after item 5): the same load on four models: a seam slice, one panel on spring edges, a seven-panel cap with membrane edge forces, and the full shell. The smallest model whose seam, frame and corner loads land within 10 percent of the full shell is the minimum section. This is Tengxiao Li's size-series idea done in the computer before in the shop.
- 2. Then, if the cap is not enough: a whole-shell model under water pressure. A closed shell reacts its own pressure, so it needs no reaction frame at all: just a pump, a gauge and a relief valve. At 1:10 it is 1.12 m across, 32 flat plates bolted at the seams, strain gauges on seams, frames and two corners. A replica sees flight stress at a given pressure, so it runs elastic at about 0.3 atm (4 kN per hex). It tests the load split, not the details. Spring 2027 at the earliest, and only if the FEA says the cap misses.

The hand numbers

Path	Hand number	Carried by
A. panel membrane	109 kN m, ~7,000 MPa	edge frame in its plane
B. seam bending	117 kN m, ~690 MPa	two frames, radial
C. seam net	1,025 kN, 126 MPa	frames in tension, corners
Per-vertex load (net)	621 kN	corner node
Bolt load across the joint	35 to 277 kN/m	depends on the split
Whole shell 1:10	run at ~0.3 atm, 4 kN/hex	self-reacting, no rig

Each number assumes its path carries everything, so they are bounds, not predictions. They are here to show that the choice matters by a factor of several, not a few percent.

This is Prof. Fleeter's point from 24 Sep: the joint is the micro scale, and the shell is the macro scale. The two have to meet in the same model before either one is final.